Tree-Level No-Drive Theorem for the Minimal Optical-Metric EM Sector in Density Field Dynamics: $\lambda_{\rm bare} = 1$ from the Pure Gauge-Invariant Action

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In Density Field Dynamics (DFD), the parameter λ introduced in Appendix R of the unified paper controls whether laboratory electromagnetic fields can pump the scalar field ψ at frequency 2ω beyond the optical-metric forward-propagation channel. Appendix R is broader than a single baseline: it contains a minimal tree-level optical-metric sector, a geometry-restoration channel, threshold-sensitive nonlinear structure, and a dual-sector κ extension. Here we address the minimal sector only. We show that in the minimal tree-level optical-metric EM sector of DFD, the linear-in-ψ EM source produced by the gauge-invariant action is proportional to the EM energy density $(E^2/c^2 + B^2)/(2\mu_0)$ rather than the stress invariant $(E^2/c^2 - B^2)/(2\mu_0)$. For ideal standing-wave cavity modes with energy equipartition, the energy-density source has no 2ω component after volume integration. Consequently the bare-coupling value of the back-reaction parameter equals unity at tree level in the minimal sector, which we write as $\lambda_{\rm bare} = 1$. The effective laboratory inference $\lambda_{\rm eff}$ differs from $\lambda_{\rm bare}$ by explicit corrections: a classical $1/Q$ mimicking effect from finite cavity quality factor, geometry-restoration contributions for non-symmetric or TE+TM superposition modes, threshold corrections if the system approaches $\eta_c = \alpha/4$, and any contributions from constitutive-sector splitting (the κ channel) or from beyond-baseline dimension-5 operators $\xi\,\psi\,F_{\mu\nu}F^{\mu\nu}$. We therefore reinterpret Appendix R's accidental bound $|\lambda - 1| < 3 \times 10^{-5}$ and the projected intentional-search reach $|\lambda - 1| \sim 10^{-14}$ as constraints on these beyond-minimal-sector channels: the minimal sector's $\lambda_{\rm bare} = 1$ is the null baseline, and nonzero inferred $\lambda_{\rm eff}$ diagnoses which beyond-baseline channel is active.

I. INTRODUCTION

Density Field Dynamics (DFD) [1] is a scalar refractive-index theory of gravity with postulates (P1) $n = e^{\psi}$ for the optical refractive index and (P2) $\mathbf{a} = (c^2/2)\nabla\psi$ for matter acceleration. These postulates specify how the scalar field ψ affects electromagnetic and material systems but leave open the inverse question: can rapidly oscillating EM fields, in turn, actively pump ψ modes beyond the standard contribution of their energy density to the matter source?

Appendix R of [1] introduces a phenomenological parameter λ to quantify this back-reaction. The parameter enters the single-mode reduction of the ψ equation of motion as the coefficient of an EM-driven source term:

$$\ddot{q} + 2\gamma_\psi \dot{q} + \Omega_\psi^2 q = \frac{(\lambda - 1)}{M_\psi} \int u(\mathbf{r})\, \Xi(\mathbf{r}, t)\, d^3r + \alpha_{\rm par} U(t) q, \tag{1}$$

where $\Xi \equiv -(1/2)e^{-2\psi_0}(B^2 - E^2/c^2)$ is the stress invariant (equivalently $-F_{\mu\nu}F^{\mu\nu}/4$ up to sign), and $\alpha_{\rm par}$ is the parametric (stiffness-modulation) coupling of Appendix R. We use the subscript "par" to distinguish it locally from the fine-structure constant α; Appendix R itself writes this coefficient simply as α. The parameter $\lambda = 1$ corresponds to no direct back-reaction drive at 2ω through this specific channel; $\lambda \neq 1$ signals an additional coupling.

The Appendix R text describes λ as "a distinct physical degree of freedom not constrained by the forward propagation relations." The accidental bound $|\lambda - 1| < 3 \times 10^{-5}$ (from the absence of parametric instability in high-Q cavities) is presented as a phenomenological constraint. Appendix R also develops additional structure beyond this minimal baseline: geometry-restoration arguments for asymmetric or TE+TM superposition modes that can restore driven overlap, threshold-sensitive nonlinear coupling activated above $\eta_c = \alpha/4$, and a dual-sector κ extension with its own leading-order phenomenology.

a. Scope of this paper. We address only the *minimal tree-level optical-metric EM sector*: the gauge-invariant action $S_{\rm EM} = -(1/4\mu_0)\int \sqrt{-\tilde{g}}\, F_{\mu\nu}F^{\mu\nu}\, d^4x$ in the DFD optical metric $n = e^{\psi}$, with no added operators and no constitutive-sector splitting. In that minimal baseline we show that the bare back-reaction parameter equals unity at tree level: $\lambda_{\rm bare} = 1$. We distinguish this from the effective laboratory inference

$$\lambda_{\rm eff} = 1 + \delta_Q + \delta_{\rm geom} + \delta_{\rm thr} + \delta_\kappa + \delta_\xi, \tag{2}$$

where $\delta_Q \sim 1/Q$ is the finite-Q equipartition-violation mimic, $\delta_{\rm geom}$ captures geometry-restoration channels for non-ideal or TE+TM modes ([1], Appendix R), $\delta_{\rm thr}$ captures threshold physics above η_c, δ_κ captures dual-sector constitutive splitting, and δ_ξ captures any beyond-baseline dimension-5 operator $\xi\,\psi\,F_{\mu\nu}F^{\mu\nu}$. The minimal-sector tree-level result $\lambda_{\rm bare} = 1$ is the null baseline; a nonzero inferred $\lambda_{\rm eff} - 1$ diagnoses which combination of these channels is active.

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Section II establishes the optical metric and EM action. Section III derives the ψ-expansion of the EM Lagrangian in a single convention. Section IV computes the cavity drive term from both (E^2+B^2) and (E^2-B^2) sources for ideal equipartitioned modes. Section V states the minimal-sector no-drive theorem. Section VI examines each beyond-baseline channel in Eq. (2) and sharpens the distinction from κ.

II. SETUP: OPTICAL METRIC AND EM ACTION

In DFD, the optical metric in Cartesian coordinates is

$$d\tilde{s}^2 = -\frac{c^2}{n^2}dt^2 + d\mathbf{x}^2, \qquad n = e^{\psi}. \tag{3}$$

The metric and inverse-metric components are

$$\tilde{g}_{00} = -c^2 e^{-2\psi}, \quad \tilde{g}_{ij} = \delta_{ij}, \tag{4}$$
$$\tilde{g}^{00} = -c^{-2} e^{2\psi}, \quad \tilde{g}^{ij} = \delta^{ij}, \tag{5}$$

with $\sqrt{-\tilde{g}} = c\, e^{-\psi}$.

The gauge-invariant EM action is

$$S_{\rm EM} = -\frac{1}{4\mu_0}\int \sqrt{-\tilde{g}}\, \tilde{g}^{\mu\alpha}\tilde{g}^{\nu\beta} F_{\mu\nu}F_{\alpha\beta}\, d^4x. \tag{6}$$

The EM postulate (P1) specifies that photons propagate along null geodesics of this metric with effective speed c/n; the action (6) is the minimal gauge-invariant action whose equations of motion reproduce this propagation.

The matter postulate (P2) couples matter to ψ through its energy density ρ as the source term in the ψ field equation. EM field energy contributes to this source through its energy-momentum content; the question is whether additional EM–ψ couplings appear beyond this energy-density channel.

III. THE ψ-EXPANSION OF THE EM LAGRANGIAN

We fix conventions once and for all: coordinates (ct, x, y, z) (time coordinate is $x^0 = ct$), metric signature $(-,+,+,+)$, $F_{0i} = E_i$, $F_{ij} = \epsilon_{ijk}B^k$ (spatial components dimensionally $[B]$), four-volume measure $d^4x = d(ct)\, d^3x$. Under these conventions, the flat-space Maxwell Lagrangian density is $\mathcal{L}^{\rm flat}_{\rm Maxwell} = (\epsilon_0/2)E^2 - (1/2\mu_0)B^2 = (1/2\mu_0)(E^2/c^2 - B^2)$, which serves as our $\psi = 0$ check. We use only these conventions throughout.

Lemma 1 (EM Lagrangian in optical metric)**.** *In the DFD optical metric (3) under the conventions above, the EM Lagrangian density $\mathcal{L}_{\rm EM}$ per unit coordinate four-volume $d^4x = d(ct)\, d^3x$ is*

$$\mathcal{L}_{\rm EM}(\psi) = \frac{1}{2\mu_0}\left[e^{\psi}\,\frac{E^2}{c^2} - e^{-\psi}\,B^2\right]. \tag{7}$$

Expanded in powers of ψ:

$$\mathcal{L}_{\rm EM}(\psi) = \frac{1}{2\mu_0}\left(\tfrac{E^2}{c^2} - B^2\right) + \frac{1}{2\mu_0}\left(\tfrac{E^2}{c^2} + B^2\right)\psi + \frac{1}{4\mu_0}\left(\tfrac{E^2}{c^2} - B^2\right)\psi^2 + O(\psi^3). \tag{8}$$

Proof. With the conventions above, compute

$$F_{\mu\nu}F^{\mu\nu} = 2F_{0i}F^{0i} + 2\sum_{i<j} F_{ij}F^{ij} = 2\tilde{g}^{00}\tilde{g}^{ij}F_{0i}F_{0j} + 2\sum_{i<j}\tilde{g}^{ik}\tilde{g}^{jl}F_{ij}F_{kl}. \tag{9}$$

Using $\tilde{g}^{00} = -c^{-2}e^{2\psi}$ and $\tilde{g}^{ij} = \delta^{ij}$, with $F_{0i} = E_i$ and $F_{ij} = \epsilon_{ijk}B^k$:

$$F_{\mu\nu}F^{\mu\nu} = -\frac{2e^{2\psi}}{c^2}E^2 + 2B^2. \tag{10}$$

The four-volume density (per $d(ct)\, d^3x$, the standard relativistic measure) is

$$\sqrt{-\tilde{g}}\Big|_{d(ct)d^3x} = \sqrt{-\tilde{g}}/c = e^{-\psi}. \tag{11}$$

(In 4D with signature $(-,+,+,+)$ and time coordinate ct, the bare $\sqrt{-\tilde{g}}$ is $c\,e^{-\psi}$; we have absorbed the factor c into the standard relativistic measure $d(ct) = c\,dt$. This is equivalent to writing the action as $\int \mathcal{L}\, d(ct)\, d^3x$ rather than $\int c\mathcal{L}\, dt\, d^3x$.) Therefore

$$\mathcal{L}_{\rm EM} = -\frac{e^{-\psi}}{4\mu_0}\left(-\frac{2e^{2\psi}E^2}{c^2} + 2B^2\right) = \frac{1}{2\mu_0}\left(\frac{e^{\psi}E^2}{c^2} - e^{-\psi}B^2\right), \tag{12}$$

as stated. Taylor expansion of $e^{\pm\psi}$ about $\psi = 0$ yields Eq. (8).

Consistency check at $\psi = 0$: $\mathcal{L}_{\rm EM}|_0 = (1/2\mu_0)(E^2/c^2 - B^2) = (\epsilon_0/2)E^2 - (1/2\mu_0)B^2$, matching the flat-space Maxwell Lagrangian density under our conventions. □

The key observation from Eq. (8): the term linear in ψ has coefficient $(E^2/c^2 + B^2)/(2\mu_0) = u_{\rm EM}$ (EM energy density), while the term quadratic in ψ has coefficient $(E^2/c^2 - B^2)/(4\mu_0)$.

Corollary 2 (Source structure for ψ from the EM sector)**.** *The EM contribution to the right-hand side of the ψ equation of motion, defined as $\partial\mathcal{L}_{\rm EM}/\partial\psi$ (the EM Lagrangian density variation with respect to ψ), is*

$$\frac{\partial\mathcal{L}_{\rm EM}}{\partial\psi}(\mathbf{r},t) = u_{\rm EM}(\mathbf{r},t) + \frac{1}{2\mu_0}\left(\tfrac{E^2}{c^2} - B^2\right)\psi(\mathbf{r},t) + O(\psi^2), \tag{13}$$

where $u_{\rm EM} = (\epsilon_0 E^2 + B^2/\mu_0)/2 = (E^2/c^2 + B^2)/(2\mu_0)$ is the EM energy density.

The full ψ equation of motion has three contributions: a gravitational-sector kinetic term, a matter-density source, and the EM contribution from $\partial\mathcal{L}_{\rm EM}/\partial\psi$. Schematically,

$$(\textit{gravitational-sector coefficient})\cdot\tilde{\Box}\psi \;=\; \rho_{\rm matter}c^2\,\psi\text{-coupling} + \frac{\partial\mathcal{L}_{\rm EM}}{\partial\psi}, \tag{14}$$

where the gravitational-sector coefficient is set by the normalization of the kinetic term in the DFD action ([1], Appendix U; it takes the form $c^4/(8\pi G)$ in Newtonian-limit conventions), and the matter coupling follows from (P2). The linearized-in-ψ EM source appearing on the right-hand side is proportional to $\partial\mathcal{L}_{\rm EM}/\partial\psi$, with an overall coefficient fixed by the gravitational-sector normalization.

The crucial structural fact for what follows is that the linear-in-ψ coefficient in Eq. (13) is proportional to $u_{\rm EM}$, not to the stress invariant $E^2/c^2 - B^2$. This proportionality is a property of $\mathcal{L}_{\rm EM}$ itself, read off directly from the expansion (8), and is independent of any choice of overall gravitational-sector normalization convention. The theorem in Section V uses only this structural proportionality; it does not require pinning down the overall coefficient.

IV. STANDING-WAVE CAVITY: WHERE DOES 2ω DRIVE COME FROM?

Consider an ideal standing-wave cavity with field profiles

$$\mathbf{E}(\mathbf{r},t) = \mathbf{E}_0(\mathbf{r})\cos(\omega t), \quad \mathbf{B}(\mathbf{r},t) = \mathbf{B}_0(\mathbf{r})\sin(\omega t), \tag{15}$$

with the 90° temporal phase shift characteristic of standing waves. The time-averaged energy equipartition condition

$$\int d^3r\,\epsilon_0|\mathbf{E}_0|^2 = \int d^3r\,\mu_0^{-1}|\mathbf{B}_0|^2 \tag{16}$$

is satisfied for any standing-wave cavity mode.

Lemma 3 (Fourier content of EM bilinears)**.** *Under the profile (15), the Fourier decomposition of E^2 and B^2 in time yields*

$$E^2(\mathbf{r},t) = \tfrac{1}{2}|\mathbf{E}_0|^2[1+\cos(2\omega t)], \tag{17}$$

$$B^2(\mathbf{r},t) = \tfrac{1}{2}|\mathbf{B}_0|^2[1-\cos(2\omega t)]. \tag{18}$$

Consequently:

$$(E^2/c^2+B^2)_{2\omega} = \tfrac{1}{2}(|\mathbf{E}_0|^2/c^2 - |\mathbf{B}_0|^2)\cos(2\omega t), \tag{19}$$

$$(E^2/c^2-B^2)_{2\omega} = \tfrac{1}{2}(|\mathbf{E}_0|^2/c^2 + |\mathbf{B}_0|^2)\cos(2\omega t). \tag{20}$$

Corollary 4 (Vanishing of energy-density drive at 2ω)**.** *For any standing-wave cavity mode with a uniform spatial test function $u(\mathbf{r}) \equiv u_0$, the integrated 2ω component of the energy-density source vanishes by equipartition:*

$$\int d^3r\,u_0\,(E^2/c^2+B^2)_{2\omega} = \frac{u_0c^2}{2}\Big(\tfrac{\langle|\mathbf{E}_0|^2\rangle}{c^2} - \langle|\mathbf{B}_0|^2\rangle\Big)\cos(2\omega t) = 0. \tag{21}$$

In contrast, the integrated (E^2/c^2-B^2) source at 2ω has its full amplitude:

$$\int d^3r\,u_0\,(E^2/c^2-B^2)_{2\omega} = u_0U_{\rm EM}\cos(2\omega t), \tag{22}$$

where $U_{\rm EM} = \int d^3r\,(\epsilon_0|\mathbf{E}_0|^2 + \mu_0^{-1}|\mathbf{B}_0|^2)/2$ is the total cavity energy.

Corollary 4 establishes the key quantitative separation: for standing-wave cavity fields, pumping at 2ω requires a source term coupled to (E^2/c^2-B^2), not to (E^2/c^2+B^2).

V. MAIN THEOREM

Theorem 5 (Tree-level no-drive theorem for the minimal optical-metric sector)**.** *Assume:*

(i) The DFD postulates P1 and P2 as in [1];
(ii) The minimal EM action Eq. (6), with no additional dimension-5 operator and no constitutive-sector splitting;
(iii) A laboratory system satisfying $\eta := U_{\rm EM}/(\rho c^2) \ll \eta_c = \alpha/4$, where ρ is the cavity-walls matter density, so that threshold physics is inactive;
(iv) A single ideal symmetric standing-wave cavity mode with exact energy equipartition ($Q\to\infty$ limit).

Then the bare-coupling value of the Appendix R back-reaction parameter is unity at tree level in the minimal sector:

$$\boxed{\lambda_{\rm bare} = 1 \quad \textit{at tree level, minimal sector}}. \tag{23}$$

Proof. By Corollary 2, the EM contribution $\partial\mathcal{L}_{\rm EM}/\partial\psi$ to the ψ equation of motion contains a zeroth-order piece $u_{\rm EM}$ (EM energy density) and a linear-in-ψ piece proportional to $(E^2/c^2-B^2)\psi$. The linear-in-ψ piece acts as a stiffness modulation (mass-like term, modifying Ω_ψ^2) rather than as a direct drive.

By Corollary 4 and hypothesis (iv), for a single ideal symmetric standing-wave cavity mode with exact equipartition, the 2ω Fourier component of the volume-integrated zeroth-order source $\int u_{\rm EM}(\mathbf{r},t)\,u(\mathbf{r})\,d^3r$ vanishes. No direct drive at frequency 2ω is produced by the minimal optical-metric action.

The Appendix R mode equation (1) identifies the drive term as $[(\lambda-1)/M_\psi]\int u(\mathbf{r})\Xi(\mathbf{r},t)\,d^3r$ with $\Xi = (1/2\mu_0)(E^2/c^2-B^2)$. Under hypotheses (i)-(iv), the minimal sector produces no such drive, so the bare-coupling value of $(\lambda-1)$ is zero at tree level in the minimal sector: $\lambda_{\rm bare} = 1$. □

We emphasize what this theorem does and does not claim. It establishes the minimal-sector tree-level baseline for an ideal symmetric single-mode standing-wave cavity. It does *not* claim that the effective laboratory inference $\lambda_{\rm eff}$ equals unity for real cavities. In particular, TE+TM superposition modes, asymmetric cavity geometries, and mode-beating configurations (all analyzed in

[1], Appendix R) can restore driven overlap even within the minimal sector; these are discussed as the $\delta_{\rm geom}$ channel in Section VI below. Additional corrections arise from each of the other beyond-baseline channels listed in the abstract.

Corollary 6 (Inferred-λ decomposition)**.** *For a laboratory experiment fitting the Appendix R mode equation (1) to cavity observations, the inferred parameter $\lambda_{\rm eff}$ decomposes as*

$$\lambda_{\rm eff} - 1 = \delta_Q + \delta_{\rm geom} + \delta_{\rm thr} + \delta_\kappa + \delta_\xi, \tag{24}$$

where $\delta_Q \sim 1/Q$ is the finite-Q equipartition-violation mimic, $\delta_{\rm geom}$ arises from geometry-restoration channels (non-ideal or TE+TM superposition modes, asymmetric cavity geometries), $\delta_{\rm thr}$ arises only if η approaches η_c, δ_κ arises from dual-sector constitutive splitting (the κ channel of [1], Appendix R), and δ_ξ arises from any beyond-baseline dimension-5 operator $\xi\,\psi\,F_{\mu\nu}F^{\mu\nu}$. Each channel is separately analyzed in Section VI and, in principle, separately constrainable by cavity-geometry, Q-scaling, threshold-approach, and constitutive-splitting experiments.

Corollary 7 (Reinterpretation of existing cavity bounds)**.** *The Appendix R accidental bound $|\lambda - 1| < 3 \times 10^{-5}$ from cavity stability is not a constraint on a single intrinsic DFD parameter but a joint constraint on $|\delta_Q + \delta_{\rm geom} + \delta_{\rm thr} + \delta_\kappa + \delta_\xi|$. Similarly the projected intentional-search reach $\sim 10^{-14}$ is a joint bound. Separating the channels requires the experimental protocols described in Section VI.*

VI. DISCUSSION

The inferred-λ decomposition Eq. (24) has five distinct channels. We analyze each in turn.

A. δ_Q: Finite-Q equipartition-violation mimic

For a real cavity with quality factor Q, equipartition of electric and magnetic energy holds only to fractional accuracy $1/Q$. The deviation produces a small residual 2ω component in the volume-integrated energy-density source $\int u\,(E^2/c^2+B^2)\,d^3r$ that was required to vanish in Corollary 4. The residual mimics a back-reaction drive:

$$\delta_Q \sim \frac{1}{Q}. \tag{25}$$

For state-of-the-art superconducting cavities with $Q \sim 10^{10}$-10^{11}, $\delta_Q \sim 10^{-10}$-10^{-11}. The existing accidental bound 3×10^{-5} is comfortably above this classical background; the projected intentional-search reach 10^{-14} is well below it, so distinguishing δ_Q from the other channels is an experimental requirement for sub-10^{-10} searches.

δ_Q is distinguishable from genuine beyond-baseline channels by Q-scaling tests: a true $\xi\psi F^2$ operator produces a Q-independent signal (up to overall amplitude), while δ_Q scales as $1/Q$.

B. $\delta_{\rm geom}$: Geometry-restoration channels

Theorem 5 covers only the ideal symmetric single-mode standing-wave case (hypothesis (iv)). Appendix R of [1] explicitly discusses geometry configurations outside that idealization which restore driven overlap:

- **TE+TM superposition modes.** A cavity excited in a mixed TE+TM mode has different spatial profiles for $|\mathbf{E}_0|^2$ and $|\mathbf{B}_0|^2$ that do not satisfy the pure equipartition structure of a single mode. The integrated 2ω component of (E^2/c^2+B^2) need not vanish for such superpositions.
- **Asymmetric cavity geometries.** Geometries with broken spatial symmetry can produce nonzero $\int u(\mathbf{r})\,(|\mathbf{E}_0|^2/c^2 - |\mathbf{B}_0|^2)(\mathbf{r})\,d^3r$ for the particular test function $u(\mathbf{r})$ of the ψ mode, even when the globally integrated equipartition holds.
- **Mode beating.** Two or more cavity modes with commensurate frequencies can produce time-varying profiles that do not individually satisfy the standing-wave assumption (ii) of Lemma 3.

Each of these restores nonzero overlap $G \equiv \int u(\mathbf{r})\,\hat{\Xi}_{2\omega}(\mathbf{r})\,d^3r$ (in the notation of Appendix R Eq. (R3)) and produces a nonzero effective drive proportional to $\delta_{\rm geom}$. We stress that these restored-overlap channels are produced by the *same* minimal optical-metric action Eq. (6): the cancellation Corollary 4 that gives $\lambda_{\rm bare} = 1$ only covered the ideal symmetric single-mode standing-wave case of hypothesis (iv), so leaving that idealization does not introduce any new operators, it only drops the specific cancellation condition. The minimal-sector action can, and in non-symmetric geometries does, produce a 2ω drive through the energy-density source without any beyond-baseline coupling being present.

Experimental isolation of $\delta_{\rm geom}$ uses cavity-geometry scans: the signal should vary with cavity shape and mode content in a calculable way, while δ_ξ (a genuine beyond-baseline operator) would not.

C. $\delta_{\rm thr}$: Threshold physics

For laboratory cavities with $\eta \sim 10^{-14} \ll \eta_c = \alpha/4 \sim 10^{-3}$, the nonlinear k_G threshold coupling ([1], Appendix G, Theorem G.2) is inactive and $\delta_{\rm thr} \approx 0$. For astrophysical plasmas where η approaches or exceeds η_c (coronal mass ejections, laboratory fusion plasmas, etc.), nonlinear activation can produce effective $\lambda \neq 1$ through the threshold channel. This is outside the hypotheses of Theorem 5.

D. δ_κ: Dual-sector constitutive splitting

This distinction is critical and deserves emphasis. Appendix R of [1] introduces a dual-sector extension with parameter κ governing electric-magnetic constitutive splitting: in that extension, the effective permittivity ϵ and permeability μ respond *differently* to ψ, rather than both being set by the single optical refractive index $n = e^\psi$. Specifically the minimal sector uses one global n for both polarizations of the EM field, while the κ extension allows distinct responses for E and B.

This is structurally different from the λ channel:

- λ controls whether a *direct* $(E^2/c^2 - B^2)\psi$ source appears in the Lagrangian, producing an inhomogeneous drive of the ψ mode.
- κ controls whether the *effective optical response* splits between E-mode and B-mode, producing a modification of the dispersion relation rather than a drive.

Both parameters are phenomenologically present in full Appendix R but are *independent* degrees of freedom. A dual-sector constitutive splitting ($\kappa \neq 0$) in general produces unequal E^2 and B^2 response spatial profiles, which breaks equipartition at the sector level and generates a nonzero effective δ_κ even in the nominally below-threshold regime. The minimal-sector theorem of this paper fixes $\lambda_{\text{bare}} = 1$ under the assumption $\kappa = 0$. A genuine measurement of $\lambda_{\text{eff}} \neq 1$ therefore does *not* automatically indicate a $\xi\psi F^2$ operator: it could instead indicate activation of the κ channel. Distinguishing the two requires independent measurement of κ through polarization-sensitive or E/B-asymmetric observables. This is a separate experimental program from the direct-drive cavity test.

The present theorem therefore does not overturn Appendix R. It establishes the null baseline for *one* of Appendix R's coupling channels, conditional on the others being inactive or held fixed.

E. δ_ξ: Beyond-baseline dimension-5 operator

Any effective Lagrangian of the form $\mathcal{L}_\xi = \xi\, \psi\, F_{\mu\nu}F^{\mu\nu}$ is gauge-invariant and general-covariant in the optical metric, and is not excluded on symmetry grounds. It is, however, *not produced* by the minimal optical-metric action Eq. (6) under the conventions of Lemma 1: that Lemma exhibits the full ψ-dependence of the minimal action, and it contains no $\psi\, F^2$ cross-term at linear order. Any nonzero ξ is therefore a beyond-baseline operator.

δ_ξ is the channel corresponding to dilaton-type coupling familiar from scalar-tensor gravity and string-theoretic dilaton models. Measuring δ_ξ separately from δ_κ, δ_Q, and δ_{geom} is a genuinely new experimental question: it would constrain the presence of a dimension-5 coupling not required by the DFD postulates P1 and P2 alone.

F. Experimental isolation protocol

To separate the five channels in Eq. (24), an experimental program would include:

1. Q-scaling series to isolate δ_Q (scales as $1/Q$, other channels do not);
2. Geometry and mode-content variation to map δ_{geom} predictions;
3. η-varying series (through cavity energy or wall density control) to test threshold behavior δ_{thr};
4. Polarization-sensitive measurements to characterize δ_κ independently;
5. Universal residual after the above subtractions, if present, constrains δ_ξ.

The minimal-sector theorem of this paper is an input to such a program, not a replacement for it.

VII. SUMMARY

We have proved a tree-level no-drive theorem for the minimal optical-metric EM sector in DFD: under DFD postulates P1 and P2, the gauge-invariant action Eq. (6) with no added operators, below-threshold conditions, ideal standing-wave equipartition, and mode symmetry, the bare-coupling value of the Appendix R back-reaction parameter is unity at tree level, $\lambda_{\text{bare}} = 1$. The derivation uses only the ψ-expansion of the minimal action and the Fourier content of standing-wave fields.

The effective laboratory inference λ_{eff} decomposes into five additive channels (Corollary 6), none of which is zero a priori in a real experiment. In particular, a nonzero $\lambda_{\text{eff}} - 1$ does not automatically diagnose beyond-DFD physics: it may reflect finite-Q equipartition violation (δ_Q), geometry-restoration channels via TE+TM superposition or cavity asymmetry (δ_{geom}), threshold physics approaching η_c (δ_{thr}), dual-sector constitutive splitting (δ_κ), or a genuine dimension-5 operator $\xi\psi F_{\mu\nu}F^{\mu\nu}$ (δ_ξ). The Appendix R accidental bound $|\lambda - 1| < 3 \times 10^{-5}$ and the projected intentional-search reach $\sim 10^{-14}$ are joint bounds on these channels. Separating them experimentally requires cavity-geometry variation, Q-scaling, threshold-approach, and polarization-sensitive protocols, outlined in Section VI.

The theorem therefore does not overturn Appendix R or eliminate λ as a phenomenological parameter. It identifies the null baseline within one specific channel and clarifies what a nonzero inferred λ_{eff} would and would not imply. This is a narrower claim than "pure DFD predicts $\lambda = 1$ exactly," which our earlier framing sometimes suggested. The narrower claim is what the minimal-sector action actually supports.

[1] G. Alcock, "Density Field Dynamics: A Complete Unified Theory," v3.4, Zenodo (2026).